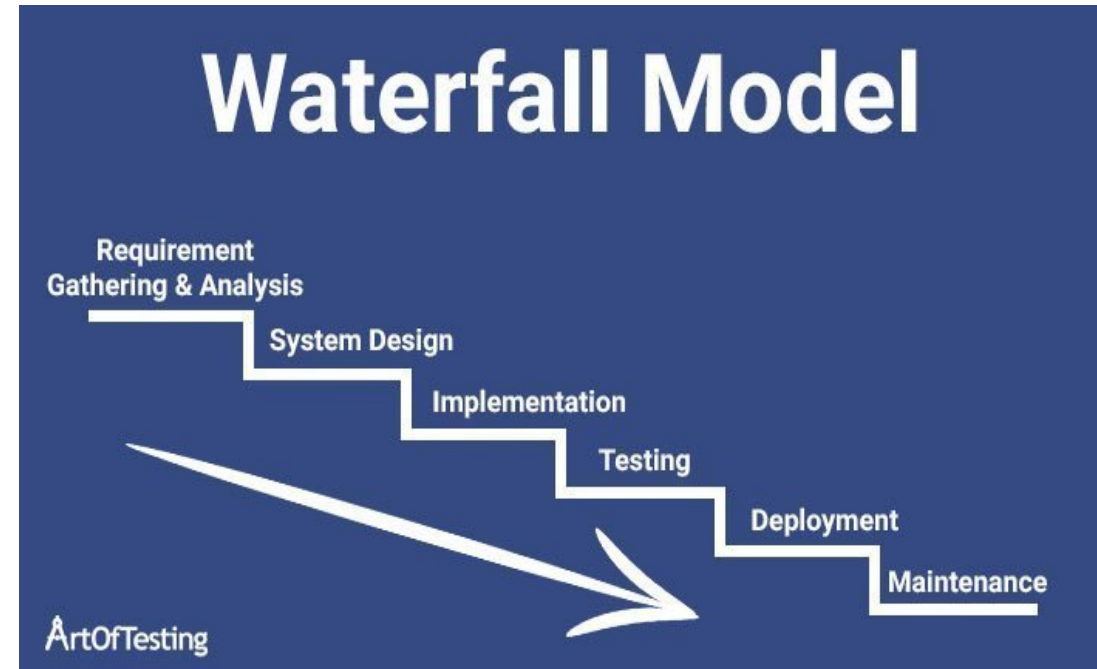


Software Development Evolution



Waterfall Model

- The waterfall model is a sequential software development methodology that follows a linear and phased approach.
- It was first introduced in the 1970s and was widely used for many years.
- The key idea behind the waterfall model is that each phase of the software development life cycle (SDLC) is completed before moving on to the next phase.



Characteristics of the Waterfall Model:

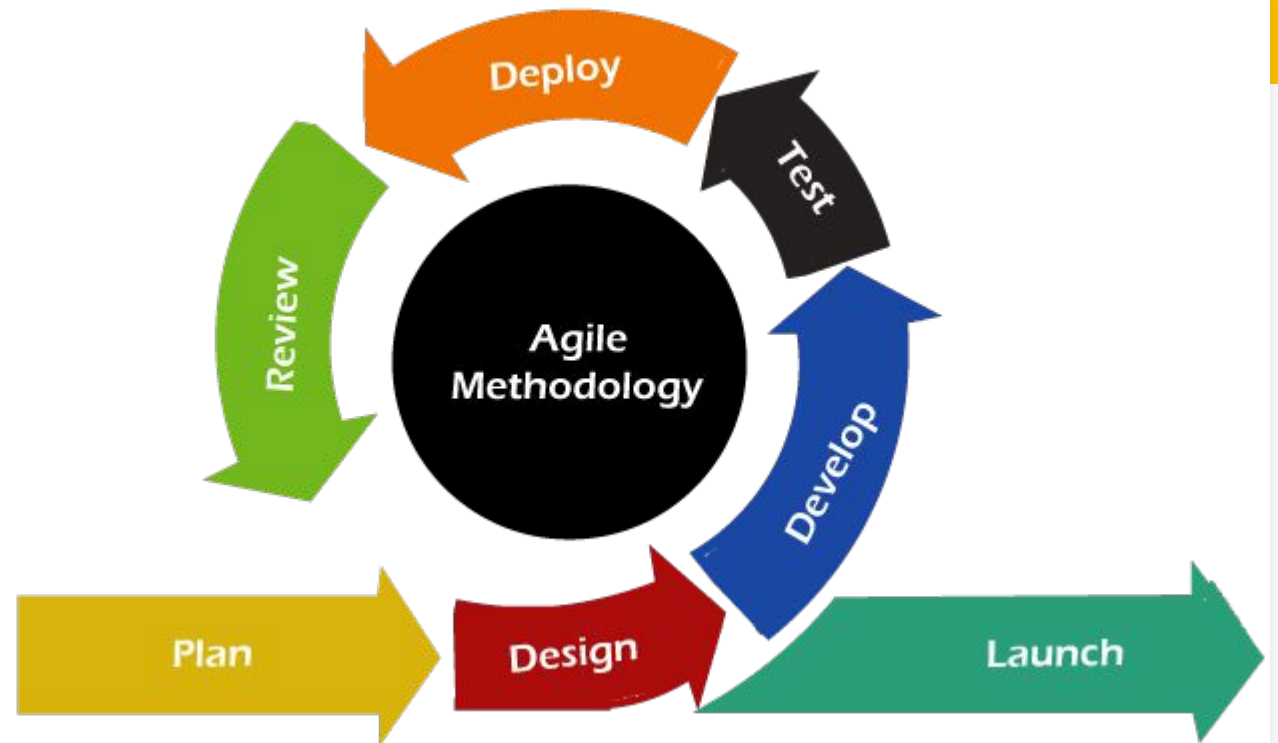
1. **Sequential Structure:** The waterfall model follows a predefined sequence of phases, typically including requirements gathering, system design, implementation, testing, deployment, and maintenance.
2. **Documentation-Driven:** The model emphasizes extensive upfront planning and documentation, with a focus on gathering and defining requirements before development begins.
3. **Clear Milestones:** Each phase in the waterfall model has distinct deliverables and milestones, making it easier to measure progress and allocate resources.
4. **Well-Suited for Stable Requirements:** The waterfall model works best when requirements are well understood, stable, and unlikely to change significantly throughout the project.
5. **Emphasis on Quality Control:** Testing is typically conducted at the end of the development process, with a dedicated testing phase to ensure the final product meets the specified requirements.

Limitations of the Waterfall Model:

1. **Lack of Flexibility:** The rigid sequential nature of the waterfall model makes it challenging to accommodate changes in requirements, as any modifications often require revisiting multiple previous phases.
2. **Limited Customer Involvement:** Stakeholder involvement and feedback are often limited to the early phases, with less opportunity for customer collaboration and validation during development.
3. **High Risk of Requirement Misalignment:** Since requirements are defined upfront and changes are difficult to accommodate, there is a higher risk of delivering a product that doesn't fully meet customer expectations.
4. **Late Detection of Issues:** Testing and quality assurance are typically performed towards the end of the development process, which can lead to late detection and costly correction of defects.
5. **Lengthy Time-to-Market:** The linear progression and dependencies between phases can lead to longer development cycles and delays in delivering a working product.

Agile Model

- The Agile model is an iterative and incremental approach to software development.
- It emphasizes flexibility, collaboration, and the ability to respond to changing requirements throughout the project.
- Agile methodologies were introduced in 2001 with the publication of the Agile Manifesto, which outlines key principles for Agile software development.



Characteristics of the Agile Model:

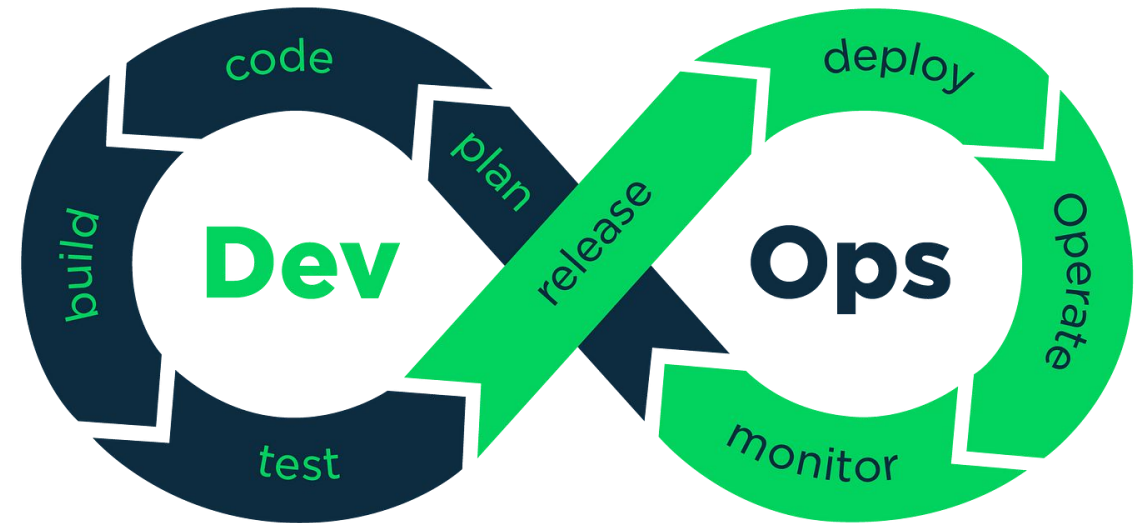
1. **Iterative Development:** Agile projects are divided into small iterations or sprints, typically lasting 1-4 weeks. Each iteration involves all phases of the software development life cycle, including planning, requirements analysis, design, development, testing, and review.
2. **Customer Collaboration:** Agile methodologies prioritize frequent customer involvement and feedback. Customers or stakeholders are actively engaged throughout the development process to provide input, review iterations, and validate the product.
3. **Adaptability and Flexibility:** Agile embraces change and allows for modifications to requirements, features, and priorities based on evolving customer needs or market conditions. Agile teams are flexible and can adjust their plans and deliverables accordingly.
4. **Self-Organizing Teams:** Agile promotes self-organizing and cross-functional teams. Team members collaborate closely, share responsibilities, and work together to deliver high-quality software.
5. **Continuous Improvement:** Agile encourages continuous learning, adaptation, and improvement. At the end of each iteration, the team reflects on its performance, identifies areas for improvement, and applies lessons learned in subsequent iterations.

Limitations of the Agile Model:

1. **Dependency on Customer Involvement:** Agile methodologies require active and continuous customer involvement. If customers or stakeholders are not available or lack clarity in their requirements, it can hinder the progress of the project.
2. **Project Scope Management:** Agile's flexibility and adaptive nature can make it challenging to manage project scope effectively. Without proper control, scope creep may occur, leading to delays and potential difficulties in meeting deadlines.
3. **Resource Allocation:** Agile teams require dedicated and skilled resources throughout the project. If resources are not readily available or allocated appropriately, it can impact the project's success.
4. **Documentation and Documentation Maintenance:** Agile methodologies prioritize working software over comprehensive documentation. While this can improve efficiency, it requires careful documentation management to ensure that essential project knowledge is captured and maintained.
5. **Resistance to Change:** Agile may face resistance from team members or stakeholders who are accustomed to more traditional and structured approaches. It requires a cultural shift and buy-in from the entire organization to fully embrace Agile practices.

DevOps

- DevOps is a software development approach that combines software development (Dev) and IT operations (Ops) to create a culture of collaboration and efficiency throughout the software delivery process.
- DevOps aims to break down silos between development and operations teams, enabling faster and more reliable software releases.



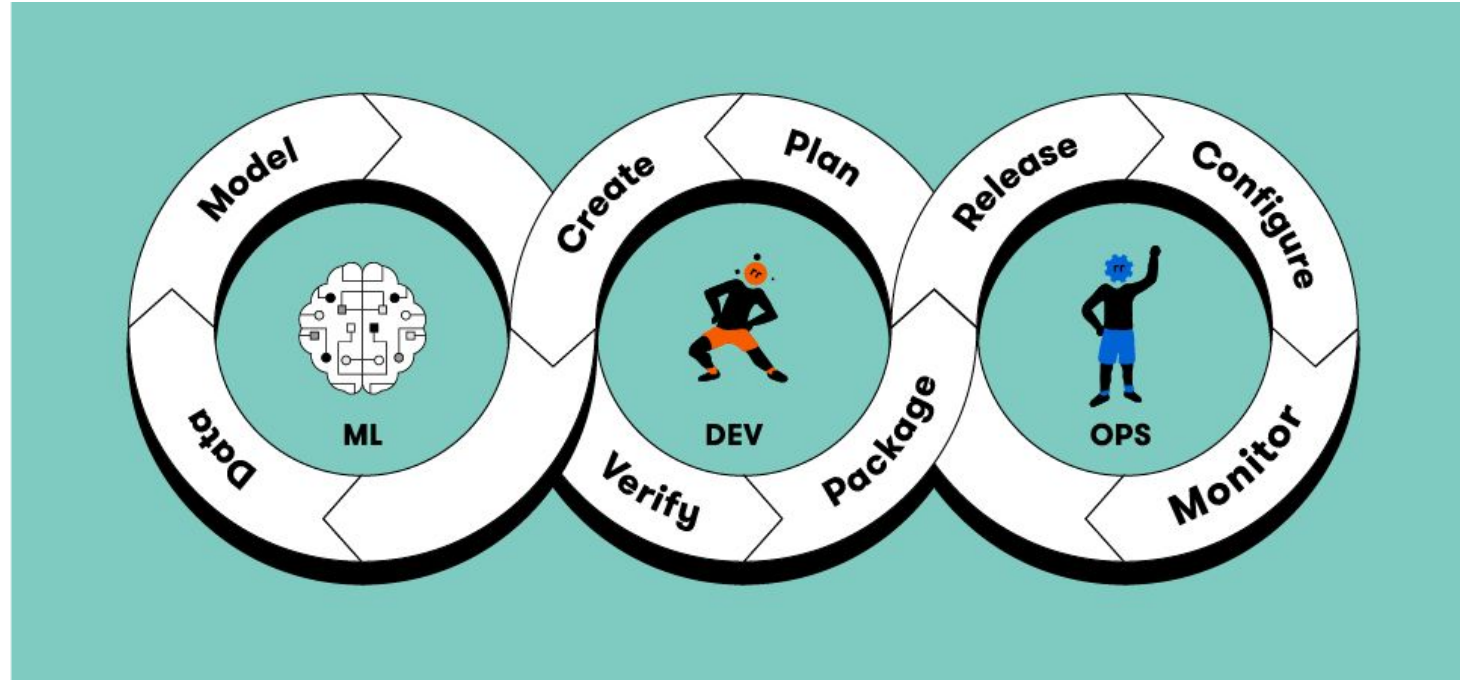
Characteristics of the DevOps Model:

1. **Collaboration and Communication:** DevOps emphasizes close collaboration and effective communication between development, operations, and other stakeholders involved in the software delivery process. This collaboration ensures that everyone is aligned and working towards shared goals.
2. **Continuous Integration and Continuous Delivery (CI/CD):** DevOps promotes the automation of software development processes, including continuous integration (merging code changes regularly) and continuous delivery (ensuring software is always in a deployable state). This automation enables faster and more frequent software releases.
3. **Infrastructure as Code:** DevOps leverages infrastructure as code (IaC) principles, enabling the management and provisioning of infrastructure resources through code. This approach ensures consistency, repeatability, and scalability in infrastructure deployments.
4. **Automation and Tooling:** DevOps encourages the use of automation and tools to streamline development, testing, deployment, and monitoring processes. These tools help automate repetitive tasks, enhance productivity, and improve the overall efficiency of the software delivery pipeline.
5. **Continuous Monitoring and Feedback:** DevOps emphasizes continuous monitoring of software performance, user feedback, and operational metrics. This feedback loop helps identify issues early on, enabling teams to respond quickly and continuously improve the software.
6. **Culture of Learning and Improvement:** DevOps fosters a culture of continuous learning, experimentation, and improvement. It encourages teams to embrace failure as an opportunity for learning, and it promotes a blameless culture where individuals feel safe to innovate and take risks.

Limitations of the DevOps Model:

1. **Cultural Transformation:** Implementing DevOps requires a significant cultural shift within organizations. It may face resistance from teams accustomed to traditional development or operations practices, and it requires strong leadership support to drive the necessary cultural changes.
2. **Complex Implementation:** DevOps implementation can be complex, especially in large organizations with legacy systems. It requires coordination and alignment across teams, as well as investment in automation tools, infrastructure, and training.
3. **Skill Set Challenges:** Adopting DevOps may require acquiring new skills or retraining team members to adapt to new practices and tools. Organizations may need to invest in training programs and provide opportunities for upskilling.
4. **Security and Compliance Considerations:** DevOps emphasizes speed and efficiency, which can potentially introduce security vulnerabilities or non-compliance risks if proper security measures and compliance controls are not integrated into the DevOps processes.
5. **Dependency on Automation:** DevOps heavily relies on automation, and if not implemented correctly or monitored adequately, it can lead to unexpected consequences or inefficiencies.

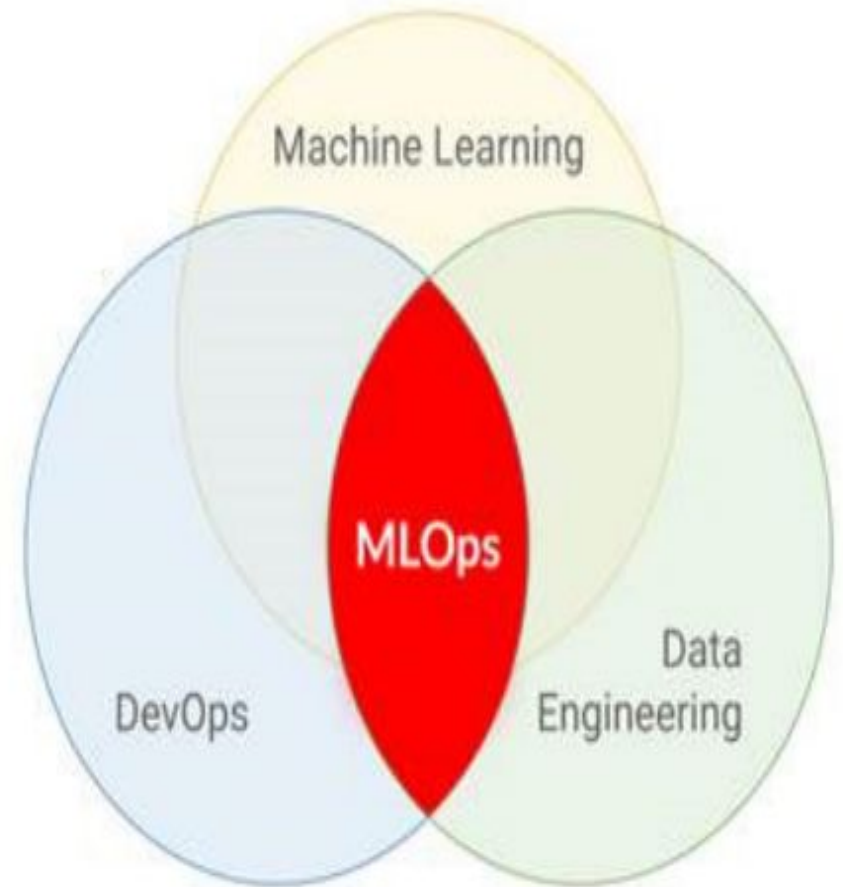
- MLOps, short for Machine Learning Operations, is the practice of managing and streamlining the lifecycle of machine learning (ML) models.
- It involves applying software engineering and DevOps principles to ML development to ensure efficient and reliable deployment, monitoring, and maintenance of ML models.



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1. Development: MLOps helps you create and train the model using data. It ensures that you have the right tools and processes in place to build accurate models.
 2. Deployment: Once the model is ready, MLOps helps you deploy it into a production environment where it can make predictions on new data. It ensures that the deployment process is smooth and reliable.
 3. Monitoring: MLOps helps you keep an eye on the model's performance. It continuously checks if the predictions are accurate and alerts you if there are any issues. This ensures that the model is working correctly and producing reliable results.
 4. Maintenance: Over time, data and requirements may change. MLOps helps you update and retrain the model when needed. It ensures that your model stays up-to-date and continues to make accurate predictions.

The goal of MLOps is to make the entire ML process more manageable, scalable, and efficient. It brings together the expertise of data scientists, software engineers, and operations teams to ensure that ML models are developed, deployed, and maintained effectively.

- Traditional software development follows a sequential approach, but machine learning (ML) development is different because it involves both code and data.
- ML models are created by applying algorithms to fit data. However, there is a disconnect between the code and data planes, which leads to challenges in deploying and maintaining ML models.
- To address these challenges, MLOps provides a systematic approach.
- It bridges the gap between code and data by bringing them together over time.
- MLOps helps streamline ML model development, deployment, and monitoring.
- It encourages collaboration between data science and IT teams, ensures accountability and traceability, and enables scalability and robustness.
- In simpler terms, MLOps is like a bridge that connects the code and data used in ML development. It helps teams work together, keeps track of their actions, and makes it easier to build and maintain ML models effectively.



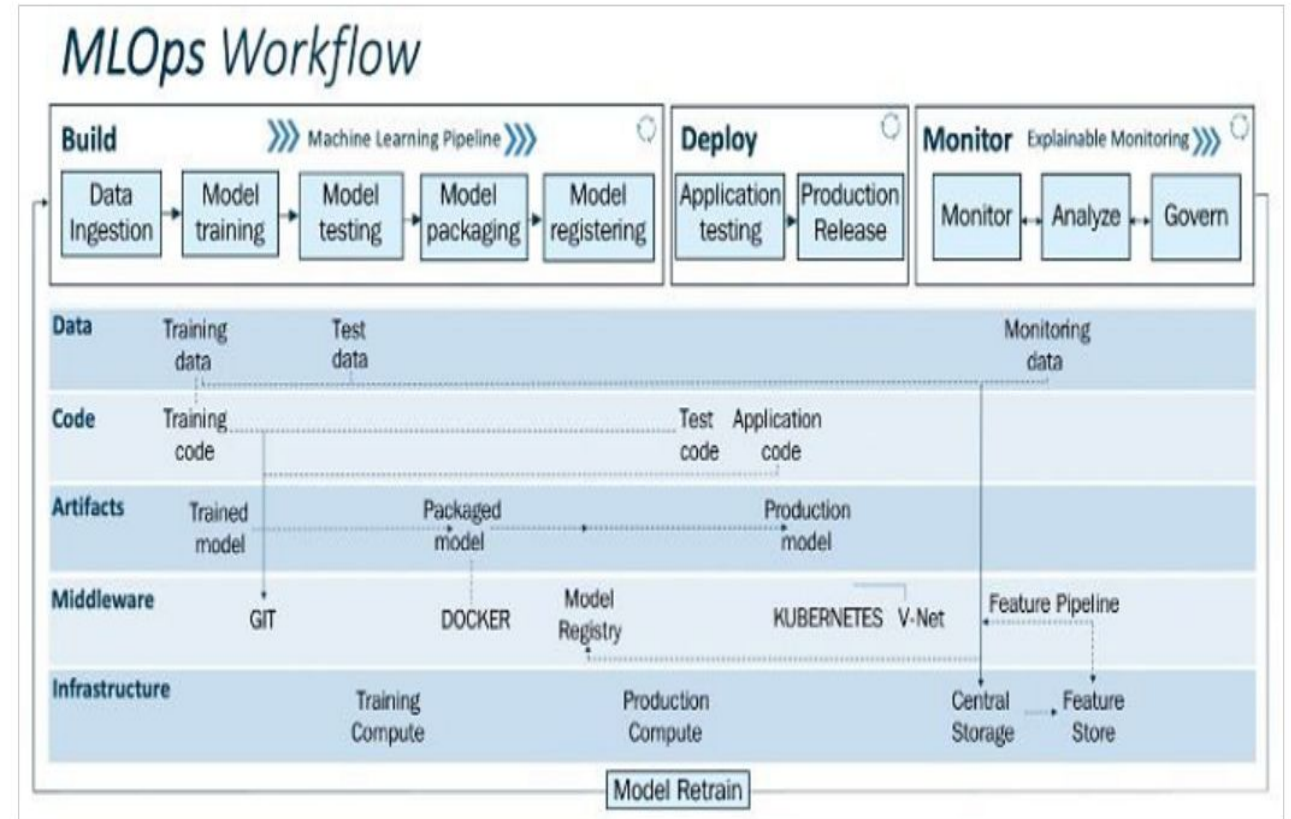
1. **Bridging Code and Data:** MLOps aims to bring code and data together over time, acknowledging their independent progression while maintaining a robust and scalable ML system. It provides mechanisms to handle the complexities of evolving data and code in a controlled manner.
2. **Challenges in ML Deployment:** ML deployment poses unique challenges, such as slow and fragmented deployment processes, lack of reproducibility, and difficulties in maintaining consistency and traceability. MLOps addresses these challenges by providing a systematic approach to model deployment, ensuring reproducibility, and enabling traceability throughout the ML lifecycle.
3. **Collaboration and Governance:** MLOps fosters collaboration between data science and IT teams. It enables cross-functional collaboration, allowing data scientists to work closely with IT professionals to validate and govern ML operations. This collaboration ensures that ML models are developed, deployed, and monitored effectively.
4. **Auditing and Traceability:** MLOps emphasizes recording and auditing all operations executed by the teams involved in ML development. This end-to-end traceability enables organizations to monitor, track, and reproduce results, ensuring accountability and compliance.
5. **Scalability and Robustness:** MLOps enables organizations to build and maintain robust and scalable ML systems. It provides mechanisms for automating repetitive tasks, managing infrastructure and resources, and implementing scalable deployment pipelines.

Concepts of MLOps:

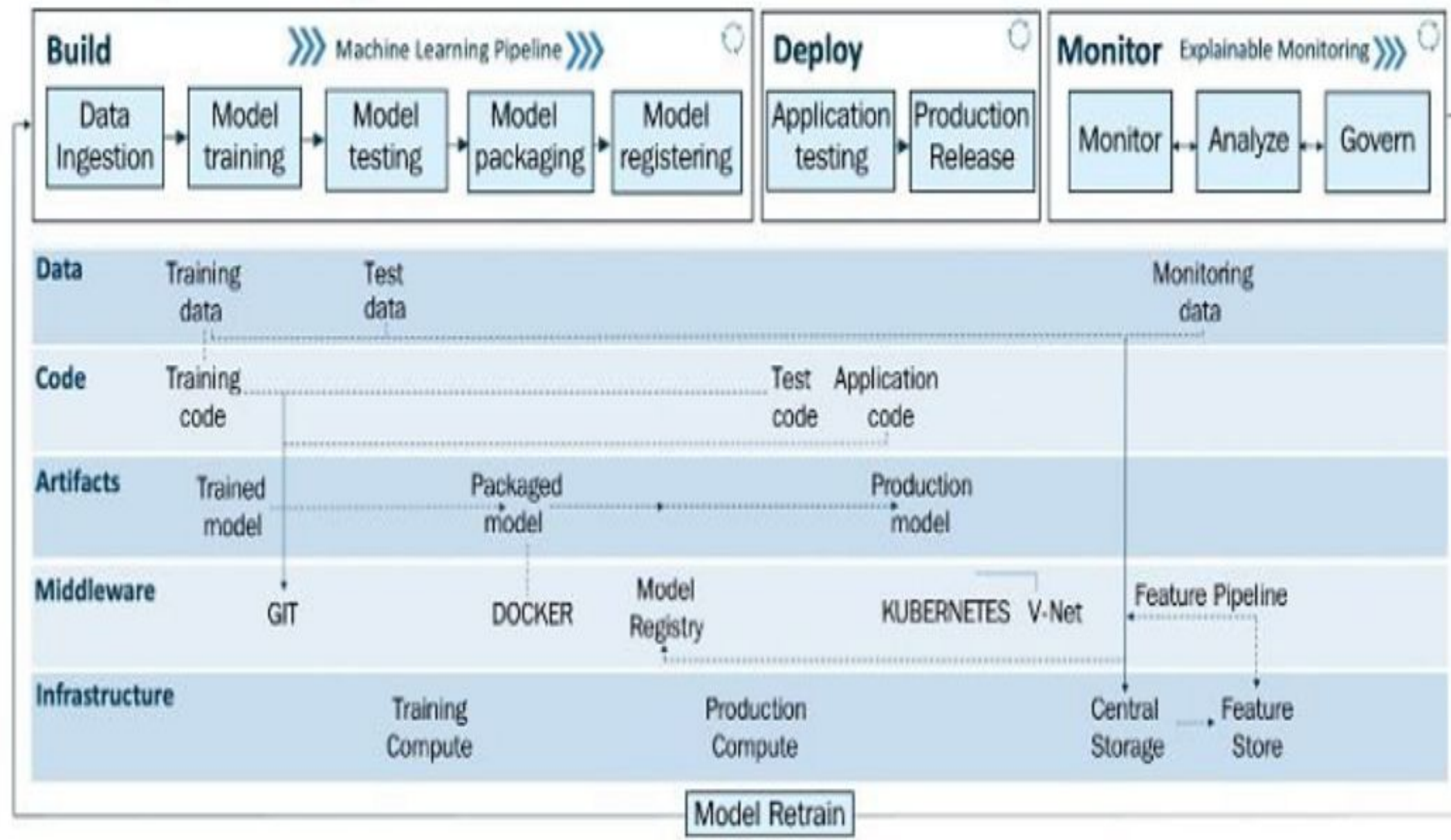
1. Collaboration: MLOps emphasizes collaboration between data scientists, ML engineers, and IT operations teams. It ensures that different stakeholders work together seamlessly throughout the ML lifecycle.
2. Automation: MLOps leverages automation to streamline repetitive tasks, such as data preprocessing, model training, deployment, and monitoring. Automation reduces manual effort, enhances efficiency, and improves consistency.
3. Version Control: MLOps applies version control techniques to track changes in ML models, code, and data. Version control enables easy collaboration, reproducibility, and traceability of experiments and model versions.
4. Continuous Integration and Continuous Deployment (CI/CD): MLOps adopts CI/CD practices to automate the integration, testing, and deployment of ML models. It ensures that models are built, validated, and deployed efficiently and consistently.

MLOps Workflow

- The upper layer (pipeline) and the lower layer (driver) of the MLOps workflow are frequently separated into two fundamental levels.
- The driver consists of data, code, artifacts, middleware, and infrastructure, whereas the pipeline consists of developing, deploying, and monitoring. The drivers down below make the pipeline possible.
- You may simply prototype, test, and validate ML models with the aid of this workflow.



MLOps Workflow



Pipeline

- The top layer is the pipeline. It is utilised for model deployment and monitoring.

1. Build Module

- This part is responsible for training, packaging and versioning part of the model.
- **Data ingestion** – It is a trigger step in the machine learning pipeline. By pulling data from diverse data sources (such as databases, data warehouses, or data lakes) and ingesting the necessary data for the model training stage, it addresses the volume, velocity, validity, and variety of data. Data validation and verification, which includes validation logic, are performed after the ingestion phase.
- **Model Training** – It contains of modular code or scripts that carry out all the usual ML operations, such as feature engineering and feature scaling, standardising, etc. To match the model to the dataset, hyper parameter adjustment is done while the ML model is being trained (training set). Manual completion of this phase is possible, however effective and automatic options include Grid Search or Random Search exist.
- **Model Testing** – Now that the model is trained and prepared, it's performance will be evaluated on the premise of its predictive outcomes on the test data. The outputs will be within the kind of a metric score.
- **Model Packaging** – After the trained model has been tested in the previous step, the model can be serialized into a file or containerized (using Docker) to be exported to the production environment.
- **Model Registration** – The model will be registered and kept in the model registry once it has been containerized in the previous stage. A registered model is composed of one or more files that construct, represent, and execute the model. The model is trained and prepared to be deployed for the production environment at this point because the full ML pipeline has been completed.

2. Deploy Module

- The operationalization of the ML models we created in the previous session is possible using the deploy module (build).
- **Testing** – Testing an ML model's performance and robustness is essential before putting it into use in production. As a result, we have the "application testing" phase where we thoroughly evaluate all the trained models for resilience and performance in a test environment that closely resembles a production environment. We deploy the models in the test environment (pre-production), which simulates the production environment, during the application testing process.
- **Release** – The previously evaluated models are now being used in production to offer useful functionality. Due to CI/CD pipelines, this production release gets deployed to the production environment.

3. Monitor Module

- The deployed ML application is tracked and evaluated in the monitor phase. Predefined metrics can be used to evaluate performance.
- **Monitor** – Data integrity, model drift, and application performance are all monitored by the monitoring module, which also gathers vital information. Telemetry information can be used to track application performance.
- It shows how a production system's device performance changed over time. The performance, health, and longevity of the production system can be monitored using telemetry data from the production system's accelerometer, gyroscope, humidity, magnetometer, pressure, and temperature sensors.
- **Analyze** – In order to increase the model's commercial relevance, model explainability approaches are used in real-time to analyse the model's fundamental properties, such as fairness, trust, bias, transparency, and error analysis.
- **Govern** – Monitoring and analysis are done to govern the deployed application to ensure that it performs at its peak for the benefit of the business (or the purpose of the ML system). We can create specific alerts and take specific measures to regulate the system after tracking and examining the production data.

4. Drivers

- **Data** – Any machine learning method can use structured or unstructured data. In ML applications, data management involves a number of procedures. Data is divided and versioned into training data, testing data, and monitoring data for the efficient operation of the ML pipeline.
- **Code** – The MLOps pipeline is powered by three key modules of code: training code, testing code, and application code. To ensure the MLOps pipeline functions properly, these scripts or pieces of code are run utilising the CI/CD and data pipelines.
- **Artifacts** – The MLOps pipeline produces artifacts including data, serialized models, code snippets, system logs, and details about the metrics used for training and testing ML models. All of these artefacts are necessary for the MLOps pipeline to operate well.
- **Middleware** – Computer software known as middleware provides functions to software programme that go beyond those provided by operating systems. Multiple apps can be used to automate and coordinate activities for the MLOps pipeline.
- **Infrastructure**– The MLOps pipeline needs critical compute and storage resources in order to develop, test, and deploy the ML models.

Business Problem: A local restaurant wants to implement a chatbot on their website to handle customer inquiries, such as providing information about the menu, operating hours, and making reservations. They believe that using a machine learning-based chatbot could improve customer experience and reduce the burden on their staff.

Solution using MLOps Workflow:

Step 1: Data Collection and Preprocessing

- Gather past customer inquiries and responses from the restaurant's website and social media platforms.
- Preprocess the data by removing any sensitive information, cleaning the text, and organizing it for training the chatbot.

Step 2: Model Development and Validation

- Data scientists develop a machine learning model for the chatbot, using natural language processing techniques to understand and respond to customer inquiries.
- Split the data into two parts: one for training the model and the other for testing its performance.

Step 3: Model Deployment

- Once the chatbot model is trained and validated, it needs to be deployed on the restaurant's website to interact with customers in real-time.

Step 4: Continuous Integration and Continuous Deployment (CI/CD)

- Set up a CI/CD pipeline to automate the process of deploying new versions of the chatbot model whenever there are updates or improvements.
- This ensures that the chatbot always uses the latest and most accurate version to provide better responses.

Step 5: Monitoring and Feedback Loop

- Implement monitoring tools to track the chatbot's interactions with customers and collect feedback.
- Regularly review customer feedback to identify areas for improvement and address any issues.

Step 6: Model Maintenance and Updates

- As the restaurant receives new inquiries and the customer's language evolves, the chatbot model may need updates to stay effective.
- Use the CI/CD pipeline to push updates to the chatbot model as needed.

Step 7: Collaboration between Data Science and IT Teams

- Foster collaboration between data science and IT teams to ensure that the chatbot deployment and maintenance processes run smoothly and efficiently.

Step 8: Auditing and Traceability

- Keep detailed records of all versions of the chatbot model, training data, and deployment changes for accountability and traceability purposes.